Re-evaluating Growth in the Time of Debt

[DSLC stages]: Data cleaning and pre-processing

1 Domain problem formulation

2 Data source overview

3 Step 1: Review background information

3.1 Data dictionary

4 Step 2: Loading in the data

```
#load the data into the project
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.6
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.2     ✓ tibble    3.3.1
✓ lubridate 1.9.4     ✓ tidyr     1.3.2
✓ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(readxl)

debt_raw <- read_excel("data/debt.xls")
growth_raw <- read_csv("data/growth.csv")
```

```
New names:
• `` -> `...3`

Warning: One or more parsing issues, call `problems()` on your data frame for details,
e.g.:
  dat <- vroom(...)
  problems(dat)
```

```
Rows: 268 Columns: 3
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (3): Data Source, World Development Indicators, ...3

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#Check the data loaded correctly
dim(debt_raw)
```

```
[1] 197 217
```

```
dim(growth_raw)
```

```
[1] 268   3
```

```
#preview the data files
head(growth_raw, 5)
```

Data Source	World Development Indicators	...3
Last Updated Date	2022-09-16	NA
Country Name	Country Code	Indicator Name,Indicator Code,1960,1961,1962,1963,1964,1965,1966,1967,1968,1969,1970,197
Aruba	ABW	GDP growth (annual %),NY.GDP.MKTP.KD.ZG,",,,,

DEBT (% of GDP)	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810	1811	1812	1813	1814	1815	1816
Albania	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data
Algeria	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data
Angola	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data

```
#remove the metadata
growth_raw <- read_csv("data/growth.csv", skip = 3)
```

```
New names:
Rows: 266 Columns: 67
── Column specification
──────────────────────────────────────────────────────── Delimiter: "," chr
(4): Country Name, Country Code, Indicator Name, Indicator Code dbl (61): 1961,
1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969, 1970, 1971, ... lgl (2): 1960,
...67
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this message.
• `` -> `...67`
```

```
#check
dim(growth_raw)
```

```
[1] 266  67
```

```
head(growth_raw, 5)
```

Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	19
Aruba	ABW	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG	NA	NA	NA	NA	NA	NA	
Africa Eastern and Southern	AFE	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG	NA	0.2395938	7.980768	5.164334	4.579952	5.327717	3.912
Afghanistan	AFG	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG	NA	NA	NA	NA	NA	NA	

Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	19
Africa Western and Central	AFW	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG	NA	1.8487517	3.770220	7.272440	5.396349	4.049793	-1.787(
Angola	AGO	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG	NA	NA	NA	NA	NA	NA	

```
#check column names match raw files
names(debt_raw)
```

```
  [1] "DEBT (% of GDP)" "1800"            "1801"            "1802"
  [5] "1803"            "1804"            "1805"            "1806"
  [9] "1807"            "1808"            "1809"            "1810"
 [13] "1811"            "1812"            "1813"            "1814"
 [17] "1815"            "1816"            "1817"            "1818"
 [21] "1819"            "1820"            "1821"            "1822"
 [25] "1823"            "1824"            "1825"            "1826"
 [29] "1827"            "1828"            "1829"            "1830"
 [33] "1831"            "1832"            "1833"            "1834"
 [37] "1835"            "1836"            "1837"            "1838"
 [41] "1839"            "1840"            "1841"            "1842"
 [45] "1843"            "1844"            "1845"            "1846"
 [49] "1847"            "1848"            "1849"            "1850"
 [53] "1851"            "1852"            "1853"            "1854"
 [57] "1855"            "1856"            "1857"            "1858"
 [61] "1859"            "1860"            "1861"            "1862"
 [65] "1863"            "1864"            "1865"            "1866"
 [69] "1867"            "1868"            "1869"            "1870"
 [73] "1871"            "1872"            "1873"            "1874"
 [77] "1875"            "1876"            "1877"            "1878"
 [81] "1879"            "1880"            "1881"            "1882"
 [85] "1883"            "1884"            "1885"            "1886"
 [89] "1887"            "1888"            "1889"            "1890"
 [93] "1891"            "1892"            "1893"            "1894"
 [97] "1895"            "1896"            "1897"            "1898"
[101] "1899"            "1900"            "1901"            "1902"
[105] "1903"            "1904"            "1905"            "1906"
[109] "1907"            "1908"            "1909"            "1910"
[113] "1911"            "1912"            "1913"            "1914"
[117] "1915"            "1916"            "1917"            "1918"
[121] "1919"            "1920"            "1921"            "1922"
[125] "1923"            "1924"            "1925"            "1926"
[129] "1927"            "1928"            "1929"            "1930"
```

```
[133] "1931"           "1932"           "1933"           "1934"
[137] "1935"           "1936"           "1937"           "1938"
[141] "1939"           "1940"           "1941"           "1942"
[145] "1943"           "1944"           "1945"           "1946"
[149] "1947"           "1948"           "1949"           "1950"
[153] "1951"           "1952"           "1953"           "1954"
[157] "1955"           "1956"           "1957"           "1958"
[161] "1959"           "1960"           "1961"           "1962"
[165] "1963"           "1964"           "1965"           "1966"
[169] "1967"           "1968"           "1969"           "1970"
[173] "1971"           "1972"           "1973"           "1974"
[177] "1975"           "1976"           "1977"           "1978"
[181] "1979"           "1980"           "1981"           "1982"
[185] "1983"           "1984"           "1985"           "1986"
[189] "1987"           "1988"           "1989"           "1990"
[193] "1991"           "1992"           "1993"           "1994"
[197] "1995"           "1996"           "1997"           "1998"
[201] "1999"           "2000"           "2001"           "2002"
[205] "2003"           "2004"           "2005"           "2006"
[209] "2007"           "2008"           "2009"           "2010"
[213] "2011"           "2012"           "2013"           "2014"
[217] "2015"
```

```
names(growth_raw)
```

```
 [1] "Country Name"   "Country Code"   "Indicator Name" "Indicator Code"
 [5] "1960"           "1961"           "1962"           "1963"
 [9] "1964"           "1965"           "1966"           "1967"
[13] "1968"           "1969"           "1970"           "1971"
[17] "1972"           "1973"           "1974"           "1975"
[21] "1976"           "1977"           "1978"           "1979"
[25] "1980"           "1981"           "1982"           "1983"
[29] "1984"           "1985"           "1986"           "1987"
[33] "1988"           "1989"           "1990"           "1991"
[37] "1992"           "1993"           "1994"           "1995"
[41] "1996"           "1997"           "1998"           "1999"
[45] "2000"           "2001"           "2002"           "2003"
[49] "2004"           "2005"           "2006"           "2007"
[53] "2008"           "2009"           "2010"           "2011"
[57] "2012"           "2013"           "2014"           "2015"
[61] "2016"           "2017"           "2018"           "2019"
[65] "2020"           "2021"           "...67"
```

Each year is an individual column for both data sources, with each column being the row observations.

5 Step 3: Examine the data and create action items

5.1 Finding invalid values

```
#examine the debt data file from the truncated years post 1960 and check for missing data inputs
debt_raw |>
  select(`1960`:`2015`) |>
  summarise(
    no_data = sum(across(everything(), ~ .x == "no data"), na.rm = TRUE)
  )
```

no_data
3419

```
#check numeric extreme values
debt_raw |>
  select(`1960`:`2015`) |>
  mutate(across(everything(), ~ as.numeric(na_if(.x, "no data")))) |>
  summarise(
    min_value = min(unlist(across(everything())), na.rm = TRUE),
    mean_value = mean(unlist(across(everything())), na.rm = TRUE),
    max_value = max(unlist(across(everything())), na.rm = TRUE)
  )
```

min_value	mean_value	max_value
0	56.10632	2092.92

The min value of 0 is expected and reasonable for our data source, 56 is a valid mean and does not need much more exploration. However the max value of 2093 does raise red flags and we must investigate further.

```
#checking the max value to see if it makes sense given our data source

debt_raw |>
  select(`DEBT (% of GDP)`, `1960`:`2015`) |>
  pivot_longer(
    cols = `1960`:`2015`,
    names_to = "year",
    values_to = "debt"
  ) |>
  mutate(debt = as.numeric(na_if(debt, "no data"))) |>
  filter(debt == max(debt, na.rm = TRUE))
```

DEBT (% of GDP)	year	debt
Nicaragua	1990	2092.92

The largest value for our debt data set containing the debts as a percentage of gdp was identified as Nicaragua in the year 1990. More research online validates this as a possible value as it is recorded that in

Nicaragua during the year 1990 was $9.74 billion USD, 33 times its good exports and more resulting in this extreme value of debt vs GDP as percentage.

This extreme value was concerning but we should also realize that it was a plausible and explainable observation and therefore should not be removed from the analysis.

We will analyse it further by examining the years surrounding 1990

```
debt_raw |>
  filter(`DEBT (% of GDP)` == "Nicaragua") |>
  select(`DEBT (% of GDP)`, `1988`:`1992`)
```

DEBT (% of GDP)	1988	1989	1990	1991	1992
Nicaragua	629.19000000000005	476.98000000000002	2092.9200000000001	333.64999999999998	448.58999999999997

While the surrounding years are not as large as in 1990, Nicaragua did have extremely large values for it recorded debt percentage potentially reinforcing our initial inclination to not remove the data point.

Doing the same quick analysis of extreme value inputs for the growth data set

```
growth_raw |>
  select(`1960`:`2021`) |>
  summarise(
    min_value = min(unlist(across(everything())), na.rm = TRUE),
    mean_value = mean(unlist(across(everything())), na.rm = TRUE),
    max_value = max(unlist(across(everything())), na.rm = TRUE)
  )
```

min_value	mean_value	max_value
-64.04711	3.675061	149.973

The initial screening of the numeric min and max values does not display any clear problems in our inuted data. Further investigating yields,

```
# examine the maximum
growth_raw |>
  select(`Country Name`, `1960`:`2021`) |>
  pivot_longer(
    cols = `1960`:`2021`,
    names_to = "year",
    values_to = "growth"
  ) |>
  filter(growth == max(growth, na.rm = TRUE))
```

Country Name	year	growth
Equatorial Guinea	1997	149.973

The maximum growth value was found to be Equatorial Guinea in 1997 at 150 pecrent year over year.

```
#examine the min
growth_raw |>
  select(`Country Name`, `1960`:`2021`) |>
  pivot_longer(
    cols = `1960`:`2021`,
    names_to = "year",
    values_to = "growth"
  ) |>
  filter(growth == min(growth, na.rm = TRUE))
```

Country Name	year	growth
Iraq	1991	-64.04711

The minimum growth value was attributed to Iran in 1991 with a growth of -64.0 percent.

Identifying both the extreme value observations we can examine the surronding years to check for any obscurities.

```
growth_raw |>
  filter(`Country Name` == "Equatorial Guinea") |>
  select(`Country Name`, `1995`:`1999`)
```

Country Name	1995	1996	1997	1998	1999
Equatorial Guinea	17.48626	66.58	149.973	23.77448	25.66402

```
growth_raw |>
  filter(`Country Name` == "Iraq") |>
  select(`Country Name`, `1989`:`1993`)
```

Country Name	1989	1990	1991	1992	1993
Iraq	-3.118535	57.81783	-64.04711	32.59221	30.28983

After checking the surrounding years I feel comfortable including the observations in our analysis, no distinct reasoning for excluding these observations.

We know move on to checking invalid and missing inputs,

```
#Check invalid inputs in growth.csv
growth_raw |>
  select(`1960`:`2021`) |>
```

```
  summarise(
    missing = sum(is.na(unlist(across(everything()))))
  )
```

missing
4069

There are 4069 missing or invalid entries in the growth data set.

```
#explicitly checking NA entries
growth_raw |>
  select(`1960`:`2021`) |>
  summarise(
    missing = sum(is.na(unlist(across(everything()))))
  )
```

missing
4069

There are the same number of NA inputs.

```
#To check if this is just the lag effect
growth_raw |>
  select(`1960`:`2021`) |>
  summarise(
    across(
      everything(),
      ~ sum(is.na(.x))
    )
  )
```

1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	1975	1976	1977	1978
266	144	139	139	139	139	131	127	126	123	126	112	112	112	112	110	106	104	98



```
#check invalid entries in debt.xls
debt_raw |>
  select(`1960`:`2015`) |>
  summarise(
    no_data = sum(across(everything(), ~ .x == "no data"), na.rm = TRUE)
  )
```

no_data
3419

```
#Count the amount of NA values explicitly
debt_raw |>
  select(`1960`:`2015`) |>
  summarise(
    missing = sum(is.na(unlist(across(everything()))))
  )
```

missing
168

There are 3419 missing values in the debt.xls file from 1960 to 2015.

These queries show that there are inconsitencies in the formatting for some of the missing values. In the debt.xls raw file from 1960 to 2015 there are 3,419 “no data” values and 168 NA values, no nonzero values, and all actual numeric inputs are valid.

There are 3419 missing values in the debt.xls file from 1960 to 2015.

We can also use the Naniar package

```
library(naniar)
```

```
Warning: package 'naniar' was built under R version 4.5.3
```

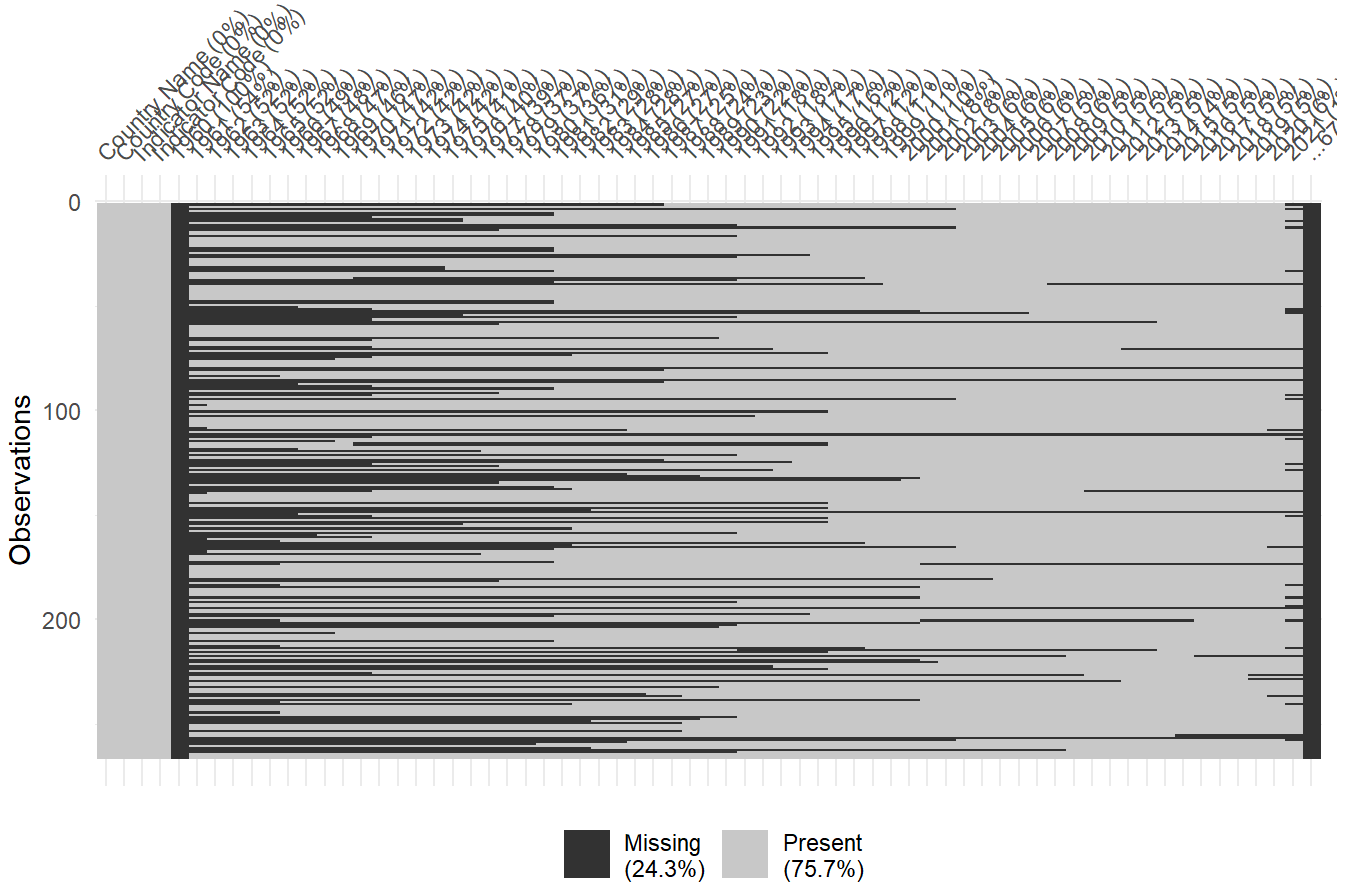
```
miss_var_summary(growth_raw)
```

variable	n_miss	pct_miss
1960	266	100
...67	266	100
1961	144	54.1
1962	139	52.3
1963	139	52.3
1964	139	52.3
1965	139	52.3
1966	131	49.2
1967	127	47.7
1968	126	47.4
1970	126	47.4
1969	123	46.2
1971	112	42.1
1972	112	42.1

variable	n_miss	pct_miss
1973	112	42.1
1974	112	42.1
1975	110	41.4
1976	106	39.8
1977	104	39.1
1978	98	36.8
1979	98	36.8
1980	97	36.5
1981	83	31.2
1982	78	29.3
1983	75	28.2
1984	75	28.2
1985	72	27.1
1986	71	26.7
1987	67	25.2
1988	64	24.1
1989	62	23.3
1990	59	22.2
1991	49	18.4
1992	48	18.0
1993	45	16.9
1994	44	16.5
1995	42	15.8
2021	38	14.3
1996	32	12.0
1997	32	12.0
1998	29	10.9
1999	28	10.5
2000	27	10.2
2001	22	8.27
2002	21	7.89
2020	17	6.39
2003	16	6.02
2004	16	6.02

variable	n_miss	pct_miss
2005	15	5.64
2006	15	5.64
2008	15	5.64
2007	14	5.26
2019	14	5.26
2009	13	4.89
2010	13	4.89
2011	13	4.89
2012	13	4.89
2013	13	4.89
2015	12	4.51
2016	12	4.51
2017	12	4.51
2018	12	4.51
2014	11	4.14
Country Name	0	0
Country Code	0	0
Indicator Name	0	0
Indicator Code	0	0

```
vis_miss(growth_raw)
```



The visualization of the missing data from the raw growth data set showed an important trend. Starting from 1960 there is a large percent of missing values that gradually decrease as we get closer to the current year. The lag effect is also observed as the number of missing values slightly increase as we get very close to the most current reservations in the data set (that being 2015 entries).

```
miss_case_summary(growth_raw)
```

case	n_miss	pct_miss
79	63	94.029851
85	63	94.029851
111	63	94.029851
148	63	94.029851
194	63	94.029851
256	63	94.029851
217	56	83.582090
57	55	82.089552
39	54	80.597015
226	54	80.597015

case	n_miss	pct_miss
229	53	79.104478
262	50	74.626866
53	49	73.134328
165	46	68.656716
180	46	68.656716
3	45	67.164179
12	45	67.164179
94	45	67.164179
257	45	67.164179
70	44	65.671642
52	43	64.179104
189	43	64.179104
220	43	64.179104
132	42	62.686567
184	42	62.686567
201	42	62.686567
219	42	62.686567
238	42	62.686567
133	41	61.194030
213	40	59.701493
163	39	58.208955
72	37	55.223881
100	37	55.223881
144	37	55.223881
146	37	55.223881
151	37	55.223881
153	37	55.223881
215	37	55.223881
223	37	55.223881
25	36	53.731343
197	36	53.731343
124	35	52.238806
128	35	52.238806
222	34	50.746269

case	n_miss	pct_miss
102	33	49.253731
11	32	47.761194
16	32	47.761194
26	32	47.761194
37	32	47.761194
55	32	47.761194
121	32	47.761194
158	32	47.761194
191	32	47.761194
202	32	47.761194
246	32	47.761194
263	32	47.761194
65	31	46.268657
203	31	46.268657
232	31	46.268657
236	31	46.268657
36	30	44.776119
131	30	44.776119
247	30	44.776119
1	29	43.283582
249	29	43.283582
253	29	43.283582
80	28	41.791045
86	28	41.791045
109	28	41.791045
115	28	41.791045
116	28	41.791045
123	28	41.791045
173	28	41.791045
235	27	40.298507
130	26	38.805970
258	26	38.805970
214	25	37.313433
138	24	35.820895

case	n_miss	pct_miss
147	24	35.820895
240	24	35.820895
248	24	35.820895
261	24	35.820895
33	23	34.328358
73	23	34.328358
137	23	34.328358
156	23	34.328358
164	23	34.328358
200	23	34.328358
5	22	32.835821
6	22	32.835821
22	22	32.835821
23	22	32.835821
38	22	32.835821
47	22	32.835821
48	22	32.835821
89	22	32.835821
136	22	32.835821
166	22	32.835821
172	22	32.835821
198	22	32.835821
210	22	32.835821
259	21	31.343284
13	19	28.358209
58	19	28.358209
91	19	28.358209
126	19	28.358209
134	19	28.358209
181	19	28.358209
9	18	26.865672
119	18	26.865672
168	18	26.865672
8	17	25.373134

case	n_miss	pct_miss
54	17	25.373134
154	17	25.373134
31	16	23.880597
32	16	23.880597
51	13	19.402985
92	13	19.402985
125	13	19.402985
150	13	19.402985
7	12	17.910448
56	12	17.910448
66	12	17.910448
69	12	17.910448
74	12	17.910448
88	12	17.910448
112	12	17.910448
160	12	17.910448
225	12	17.910448
75	10	14.925373
114	10	14.925373
206	10	14.925373
159	9	13.432836
255	9	13.432836
50	8	11.940299
87	8	11.940299
118	8	11.940299
149	8	11.940299
183	8	11.940299
83	7	10.447761
162	7	10.447761
212	7	10.447761
239	7	10.447761
244	7	10.447761
228	5	7.462687
97	3	4.477612

case	n_miss	pct_miss
108	3	4.477612
113	3	4.477612
139	3	4.477612
161	3	4.477612
167	3	4.477612
193	3	4.477612
2	2	2.985075
4	2	2.985075
10	2	2.985075
14	2	2.985075
15	2	2.985075
17	2	2.985075
18	2	2.985075
19	2	2.985075
20	2	2.985075
21	2	2.985075
24	2	2.985075
27	2	2.985075
28	2	2.985075
29	2	2.985075
30	2	2.985075
34	2	2.985075
35	2	2.985075
40	2	2.985075
41	2	2.985075
42	2	2.985075
43	2	2.985075
44	2	2.985075
45	2	2.985075
46	2	2.985075
49	2	2.985075
59	2	2.985075
60	2	2.985075
61	2	2.985075

case	n_miss	pct_miss
62	2	2.985075
63	2	2.985075
64	2	2.985075
67	2	2.985075
68	2	2.985075
71	2	2.985075
76	2	2.985075
77	2	2.985075
78	2	2.985075
81	2	2.985075
82	2	2.985075
84	2	2.985075
90	2	2.985075
93	2	2.985075
95	2	2.985075
96	2	2.985075
98	2	2.985075
99	2	2.985075
101	2	2.985075
103	2	2.985075
104	2	2.985075
105	2	2.985075
106	2	2.985075
107	2	2.985075
110	2	2.985075
117	2	2.985075
120	2	2.985075
122	2	2.985075
127	2	2.985075
129	2	2.985075
135	2	2.985075
140	2	2.985075
141	2	2.985075
142	2	2.985075

case	n_miss	pct_miss
143	2	2.985075
145	2	2.985075
152	2	2.985075
155	2	2.985075
157	2	2.985075
169	2	2.985075
170	2	2.985075
171	2	2.985075
174	2	2.985075
175	2	2.985075
176	2	2.985075
177	2	2.985075
178	2	2.985075
179	2	2.985075
182	2	2.985075
185	2	2.985075
186	2	2.985075
187	2	2.985075
188	2	2.985075
190	2	2.985075
192	2	2.985075
195	2	2.985075
196	2	2.985075
199	2	2.985075
204	2	2.985075
205	2	2.985075
207	2	2.985075
208	2	2.985075
209	2	2.985075
211	2	2.985075
216	2	2.985075
218	2	2.985075
221	2	2.985075
224	2	2.985075

case	n_miss	pct_miss
227	2	2.985075
230	2	2.985075
231	2	2.985075
233	2	2.985075
234	2	2.985075
237	2	2.985075
241	2	2.985075
242	2	2.985075
243	2	2.985075
245	2	2.985075
250	2	2.985075
251	2	2.985075
252	2	2.985075
254	2	2.985075
260	2	2.985075
264	2	2.985075
265	2	2.985075
266	2	2.985075

```
growth_raw |>
  mutate(row = row_number()) |>
  filter(row %in% c(79, 85, 111, 148, 194, 256)) |>
  select(row, `Country Name`, `Country Code`)
```

row	Country Name	Country Code
79	Faroe Islands	FRO
85	Gibraltar	GIB
111	Not classified	INX
148	St. Martin (French part)	MAF
194	Korea, Dem. People's Rep.	PRK
256	British Virgin Islands	VGB

Some countries have very low entries overall for example the Faroe Islands, Gibraltar, St. Martin... Some countries have 0 observations overall

```
growth_raw |>
  summarise(
```

```
    missing = sum(is.na(`...67`)),
    total = n()
  )
```

missing	total
266	266

Therefore since there are no observations for 67 we should just remove this from the analyse.

We can also do the same checks for the debt data file as well.

```
miss_var_summary(debt_raw) |>
  arrange(desc(n_miss))
```

variable	n_miss	pct_miss
1800	3	1.52
1801	3	1.52
1802	3	1.52
1803	3	1.52
1804	3	1.52
1805	3	1.52
1806	3	1.52
1807	3	1.52
1808	3	1.52
1809	3	1.52
1810	3	1.52
1811	3	1.52
1812	3	1.52
1813	3	1.52
1814	3	1.52
1815	3	1.52
1816	3	1.52
1817	3	1.52
1818	3	1.52
1819	3	1.52
1820	3	1.52
1821	3	1.52
1822	3	1.52

variable	n_miss	pct_miss
1823	3	1.52
1824	3	1.52
1825	3	1.52
1826	3	1.52
1827	3	1.52
1828	3	1.52
1829	3	1.52
1830	3	1.52
1831	3	1.52
1832	3	1.52
1833	3	1.52
1834	3	1.52
1835	3	1.52
1836	3	1.52
1837	3	1.52
1838	3	1.52
1839	3	1.52
1840	3	1.52
1841	3	1.52
1842	3	1.52
1843	3	1.52
1844	3	1.52
1845	3	1.52
1846	3	1.52
1847	3	1.52
1848	3	1.52
1849	3	1.52
1850	3	1.52
1851	3	1.52
1852	3	1.52
1853	3	1.52
1854	3	1.52
1855	3	1.52
1856	3	1.52

variable	n_miss	pct_miss
1857	3	1.52
1858	3	1.52
1859	3	1.52
1860	3	1.52
1861	3	1.52
1862	3	1.52
1863	3	1.52
1864	3	1.52
1865	3	1.52
1866	3	1.52
1867	3	1.52
1868	3	1.52
1869	3	1.52
1870	3	1.52
1871	3	1.52
1872	3	1.52
1873	3	1.52
1874	3	1.52
1875	3	1.52
1876	3	1.52
1877	3	1.52
1878	3	1.52
1879	3	1.52
1880	3	1.52
1881	3	1.52
1882	3	1.52
1883	3	1.52
1884	3	1.52
1885	3	1.52
1886	3	1.52
1887	3	1.52
1888	3	1.52
1889	3	1.52
1890	3	1.52

variable	n_miss	pct_miss
1891	3	1.52
1892	3	1.52
1893	3	1.52
1894	3	1.52
1895	3	1.52
1896	3	1.52
1897	3	1.52
1898	3	1.52
1899	3	1.52
1900	3	1.52
1901	3	1.52
1902	3	1.52
1903	3	1.52
1904	3	1.52
1905	3	1.52
1906	3	1.52
1907	3	1.52
1908	3	1.52
1909	3	1.52
1910	3	1.52
1911	3	1.52
1912	3	1.52
1913	3	1.52
1914	3	1.52
1915	3	1.52
1916	3	1.52
1917	3	1.52
1918	3	1.52
1919	3	1.52
1920	3	1.52
1921	3	1.52
1922	3	1.52
1923	3	1.52
1924	3	1.52

variable	n_miss	pct_miss
1925	3	1.52
1926	3	1.52
1927	3	1.52
1928	3	1.52
1929	3	1.52
1930	3	1.52
1931	3	1.52
1932	3	1.52
1933	3	1.52
1934	3	1.52
1935	3	1.52
1936	3	1.52
1937	3	1.52
1938	3	1.52
1939	3	1.52
1940	3	1.52
1941	3	1.52
1942	3	1.52
1943	3	1.52
1944	3	1.52
1945	3	1.52
1946	3	1.52
1947	3	1.52
1948	3	1.52
1949	3	1.52
1950	3	1.52
1951	3	1.52
1952	3	1.52
1953	3	1.52
1954	3	1.52
1955	3	1.52
1956	3	1.52
1957	3	1.52
1958	3	1.52

variable	n_miss	pct_miss
1959	3	1.52
1960	3	1.52
1961	3	1.52
1962	3	1.52
1963	3	1.52
1964	3	1.52
1965	3	1.52
1966	3	1.52
1967	3	1.52
1968	3	1.52
1969	3	1.52
1970	3	1.52
1971	3	1.52
1972	3	1.52
1973	3	1.52
1974	3	1.52
1975	3	1.52
1976	3	1.52
1977	3	1.52
1978	3	1.52
1979	3	1.52
1980	3	1.52
1981	3	1.52
1982	3	1.52
1983	3	1.52
1984	3	1.52
1985	3	1.52
1986	3	1.52
1987	3	1.52
1988	3	1.52
1989	3	1.52
1990	3	1.52
1991	3	1.52
1992	3	1.52

variable	n_miss	pct_miss
1993	3	1.52
1994	3	1.52
1995	3	1.52
1996	3	1.52
1997	3	1.52
1998	3	1.52
1999	3	1.52
2000	3	1.52
2001	3	1.52
2002	3	1.52
2003	3	1.52
2004	3	1.52
2005	3	1.52
2006	3	1.52
2007	3	1.52
2008	3	1.52
2009	3	1.52
2010	3	1.52
2011	3	1.52
2012	3	1.52
2013	3	1.52
2014	3	1.52
2015	3	1.52
DEBT (% of GDP)	2	1.02

```
#identify culprits
miss_case_summary(debt_raw) |>
  arrange(desc(n_miss))
```

case	n_miss	pct_miss
1	217	100.00000
196	217	100.00000
197	216	99.53917
2	0	0.00000
3	0	0.00000

case	n_miss	pct_miss
4	0	0.00000
5	0	0.00000
6	0	0.00000
7	0	0.00000
8	0	0.00000
9	0	0.00000
10	0	0.00000
11	0	0.00000
12	0	0.00000
13	0	0.00000
14	0	0.00000
15	0	0.00000
16	0	0.00000
17	0	0.00000
18	0	0.00000
19	0	0.00000
20	0	0.00000
21	0	0.00000
22	0	0.00000
23	0	0.00000
24	0	0.00000
25	0	0.00000
26	0	0.00000
27	0	0.00000
28	0	0.00000
29	0	0.00000
30	0	0.00000
31	0	0.00000
32	0	0.00000
33	0	0.00000
34	0	0.00000
35	0	0.00000
36	0	0.00000
37	0	0.00000

case	n_miss	pct_miss
38	0	0.00000
39	0	0.00000
40	0	0.00000
41	0	0.00000
42	0	0.00000
43	0	0.00000
44	0	0.00000
45	0	0.00000
46	0	0.00000
47	0	0.00000
48	0	0.00000
49	0	0.00000
50	0	0.00000
51	0	0.00000
52	0	0.00000
53	0	0.00000
54	0	0.00000
55	0	0.00000
56	0	0.00000
57	0	0.00000
58	0	0.00000
59	0	0.00000
60	0	0.00000
61	0	0.00000
62	0	0.00000
63	0	0.00000
64	0	0.00000
65	0	0.00000
66	0	0.00000
67	0	0.00000
68	0	0.00000
69	0	0.00000
70	0	0.00000
71	0	0.00000

case	n_miss	pct_miss
72	0	0.00000
73	0	0.00000
74	0	0.00000
75	0	0.00000
76	0	0.00000
77	0	0.00000
78	0	0.00000
79	0	0.00000
80	0	0.00000
81	0	0.00000
82	0	0.00000
83	0	0.00000
84	0	0.00000
85	0	0.00000
86	0	0.00000
87	0	0.00000
88	0	0.00000
89	0	0.00000
90	0	0.00000
91	0	0.00000
92	0	0.00000
93	0	0.00000
94	0	0.00000
95	0	0.00000
96	0	0.00000
97	0	0.00000
98	0	0.00000
99	0	0.00000
100	0	0.00000
101	0	0.00000
102	0	0.00000
103	0	0.00000
104	0	0.00000
105	0	0.00000

case	n_miss	pct_miss
106	0	0.00000
107	0	0.00000
108	0	0.00000
109	0	0.00000
110	0	0.00000
111	0	0.00000
112	0	0.00000
113	0	0.00000
114	0	0.00000
115	0	0.00000
116	0	0.00000
117	0	0.00000
118	0	0.00000
119	0	0.00000
120	0	0.00000
121	0	0.00000
122	0	0.00000
123	0	0.00000
124	0	0.00000
125	0	0.00000
126	0	0.00000
127	0	0.00000
128	0	0.00000
129	0	0.00000
130	0	0.00000
131	0	0.00000
132	0	0.00000
133	0	0.00000
134	0	0.00000
135	0	0.00000
136	0	0.00000
137	0	0.00000
138	0	0.00000
139	0	0.00000

case	n_miss	pct_miss
140	0	0.00000
141	0	0.00000
142	0	0.00000
143	0	0.00000
144	0	0.00000
145	0	0.00000
146	0	0.00000
147	0	0.00000
148	0	0.00000
149	0	0.00000
150	0	0.00000
151	0	0.00000
152	0	0.00000
153	0	0.00000
154	0	0.00000
155	0	0.00000
156	0	0.00000
157	0	0.00000
158	0	0.00000
159	0	0.00000
160	0	0.00000
161	0	0.00000
162	0	0.00000
163	0	0.00000
164	0	0.00000
165	0	0.00000
166	0	0.00000
167	0	0.00000
168	0	0.00000
169	0	0.00000
170	0	0.00000
171	0	0.00000
172	0	0.00000
173	0	0.00000

case	n_miss	pct_miss
174	0	0.00000
175	0	0.00000
176	0	0.00000
177	0	0.00000
178	0	0.00000
179	0	0.00000
180	0	0.00000
181	0	0.00000
182	0	0.00000
183	0	0.00000
184	0	0.00000
185	0	0.00000
186	0	0.00000
187	0	0.00000
188	0	0.00000
189	0	0.00000
190	0	0.00000
191	0	0.00000
192	0	0.00000
193	0	0.00000
194	0	0.00000
195	0	0.00000

There are 3 rows completely empty

```
debt_raw |>
  mutate(row = row_number()) |>
  filter(row %in% c(1, 196, 197)) |>
  select(row, `DEBT (% of GDP)`)
```

row	DEBT (% of GDP)
1	NA
196	NA
197	©IMF, 2019

```
debt_raw |>
  select(`DEBT (% of GDP)`, `1960`:`1965`) |>
```

```
  slice(c(1:5, 193:197))
```

DEBT (% of GDP)	1960	1961	1962	1963	1964
NA	NA	NA	NA	NA	NA
Afghanistan	no data	no data	no data	no data	no data
Albania	no data	no data	no data	no data	no data
Algeria	no data	no data	no data	no data	no data
Angola	no data	no data	no data	no data	no data
G-20 Advanced	31.43	30.02	28.960000000000001	27.170000000000002	27.21000000000000
G-20 Emerging	18.309999999999999	25.370000000000001	32.509999999999998	35.490000000000002	30.35000000000000
Low Income	8.9700000000000006	8.2200000000000006	10.16	30.510000000000002	24.46999999999999
NA	NA	NA	NA	NA	NA
©IMF, 2019	NA	NA	NA	NA	NA

Some of the entries are not countries explicitly but rather conglomerates or metadata rows, which should most likely be removed.

```
debt_raw |>
  select(`DEBT (% of GDP)`) |>
  print(n = Inf)
```

```
# A tibble: 197 × 1
    `DEBT (% of GDP)`
    <chr>
  1 <NA>
  2 Afghanistan
  3 Albania
  4 Algeria
  5 Angola
  6 Anguilla
  7 Antigua and Barbuda
  8 Argentina
  9 Armenia
 10 Australia
 11 Austria
 12 Azerbaijan
 13 Bahamas, The
 14 Bahrain
 15 Bangladesh
 16 Barbados
```

17 Belarus
18 Belgium
19 Belize
20 Benin
21 Bhutan
22 Bolivia
23 Bosnia and Herzegovina
24 Botswana
25 Brazil
26 Brunei Darussalam
27 Bulgaria
28 Burkina Faso
29 Burundi
30 Cabo Verde
31 Cambodia
32 Cameroon
33 Canada
34 Central African Republic
35 Chad
36 Chile
37 China, People's Republic of
38 Colombia
39 Comoros
40 Congo, Dem. Rep. of the
41 Congo, Republic of
42 Costa Rica
43 Croatia
44 Cyprus
45 Czech Republic
46 Côte d'Ivoire
47 Denmark
48 Djibouti
49 Dominica
50 Dominican Republic
51 Ecuador
52 Egypt
53 El Salvador
54 Equatorial Guinea
55 Eritrea
56 Estonia
57 Eswatini
58 Ethiopia
59 Fiji
60 Finland
61 France
62 Gabon
63 Gambia, The
64 Georgia
65 Germany
66 Ghana
67 Greece

Grenada
Guatemala
Guinea
Guinea-Bissau
Guyana
Haiti
Honduras
Hong Kong SAR
Hungary
Iceland
India
Indonesia
Iran
Iraq
Ireland
Israel
Italy
Jamaica
Japan
Jordan
Kazakhstan
Kenya
Kiribati
Korea, Republic of
Kosovo
Kuwait
Kyrgyz Republic
Lao P.D.R.
Latvia
Lebanon
Lesotho
Liberia
Libya
Lithuania
Luxembourg
Madagascar
Malawi
Malaysia
Maldives
Mali
Malta
Marshall Islands
Mauritania
Mauritius
Mexico
Micronesia, Fed. States of
Moldova
Mongolia
Montenegro
Morocco
Mozambique

119 Myanmar
120 Namibia
121 Nepal
122 Netherlands
123 New Zealand
124 Nicaragua
125 Niger
126 Nigeria
127 North Macedonia
128 Norway
129 Oman
130 Pakistan
131 Panama
132 Papua New Guinea
133 Paraguay
134 Peru
135 Philippines
136 Poland
137 Portugal
138 Puerto Rico
139 Qatar
140 Romania
141 Russian Federation
142 Rwanda
143 Saint Kitts and Nevis
144 Saint Lucia
145 Saint Vincent and the Grenadines
146 Samoa
147 San Marino
148 Saudi Arabia
149 Senegal
150 Serbia
151 Seychelles
152 Sierra Leone
153 Singapore
154 Slovak Republic
155 Slovenia
156 Solomon Islands
157 South Africa
158 South Sudan, Republic of
159 Spain
160 Sri Lanka
161 Sudan
162 Suriname
163 Sweden
164 Switzerland
165 Syria
166 São Tomé and Príncipe
167 Taiwan Province of China
168 Tajikistan
169 Tanzania

```
170 Thailand
171 Togo
172 Tonga
173 Trinidad and Tobago
174 Tunisia
175 Turkey
176 Turkmenistan
177 Tuvalu
178 Uganda
179 Ukraine
180 United Arab Emirates
181 United Kingdom
182 United States
183 Uruguay
184 Uzbekistan
185 Vanuatu
186 Venezuela
187 Vietnam
188 Yemen
189 Zambia
190 Zimbabwe
191 Eastern Caribbean Currency Union
192 Euro area
193 G-20 Advanced
194 G-20 Emerging
195 Low Income
196 <NA>
197 ©IMF, 2019
```

We should only keep rows 2-180 which matches the original statement that the data set contains information of 189 countries. But now we must also check the growth data set for th same inconsistencies.

```
growth_raw |>
  select(`Country Name`, `Country Code`) |>
  tail(30)
```

Country Name	Country Code
Latin America & the Caribbean (IDA & IBRD countries)	TLA
Timor-Leste	TLS
Middle East & North Africa (IDA & IBRD countries)	TMN
Tonga	TON
South Asia (IDA & IBRD)	TSA
Sub-Saharan Africa (IDA & IBRD countries)	TSS
Trinidad and Tobago	TTO
Tunisia	TUN

Country Name	Country Code
Turkiye	TUR
Tuvalu	TUV
Tanzania	TZA
Uganda	UGA
Ukraine	UKR
Upper middle income	UMC
Uruguay	URY
United States	USA
Uzbekistan	UZB
St. Vincent and the Grenadines	VCT
Venezuela, RB	VEN
British Virgin Islands	VGB
Virgin Islands (U.S.)	VIR
Vietnam	VNM
Vanuatu	VUT
World	WLD
Samoa	WSM
Kosovo	XKX
Yemen, Rep.	YEM
South Africa	ZAF
Zambia	ZMB
Zimbabwe	ZWE

```
growth_raw |>
  select(`Country Name`, `Country Code`) |>
  tail(30) |>
  print(n = 30)
```

```
# A tibble: 30 × 2
   `Country Name`                                      `Country Code`
   <chr>                                               <chr>
 1 Latin America & the Caribbean (IDA & IBRD countries) TLA
 2 Timor-Leste                                         TLS
 3 Middle East & North Africa (IDA & IBRD countries)   TMN
 4 Tonga                                               TON
 5 South Asia (IDA & IBRD)                             TSA
 6 Sub-Saharan Africa (IDA & IBRD countries)           TSS
 7 Trinidad and Tobago                                 TTO
 8 Tunisia                                             TUN
```

```
 9 Turkiye                                  TUR
10 Tuvalu                                   TUV
11 Tanzania                                 TZA
12 Uganda                                   UGA
13 Ukraine                                  UKR
14 Upper middle income                      UMC
15 Uruguay                                  URY
16 United States                            USA
17 Uzbekistan                               UZB
18 St. Vincent and the Grenadines           VCT
19 Venezuela, RB                            VEN
20 British Virgin Islands                   VGB
21 Virgin Islands (U.S.)                    VIR
22 Vietnam                                  VNM
23 Vanuatu                                  VUT
24 World                                    WLD
25 Samoa                                    WSM
26 Kosovo                                   XKX
27 Yemen, Rep.                              YEM
28 South Africa                             ZAF
29 Zambia                                   ZMB
30 Zimbabwe                                 ZWE
```

```
growth_raw |>
  filter(
    str_detect(
      `Country Name`,
      "World|income|IDA|IBRD|Africa|Asia|America|Europe|Caribbean|Pacific|Middle East"
    )
  ) |>
  select(`Country Name`, `Country Code`)
```

Country Name	Country Code
Africa Eastern and Southern	AFE
Africa Western and Central	AFW
Arab World	ARB
American Samoa	ASM
Central African Republic	CAF
Central Europe and the Baltics	CEB
Caribbean small states	CSS
East Asia & Pacific (excluding high income)	EAP
East Asia & Pacific	EAS
Europe & Central Asia (excluding high income)	ECA
Europe & Central Asia	ECS

Country Name	Country Code
European Union	EUU
High income	HIC
IBRD only	IBD
IDA & IBRD total	IBT
IDA total	IDA
IDA blend	IDB
IDA only	IDX
Latin America & Caribbean (excluding high income)	LAC
Latin America & Caribbean	LCN
Low income	LIC
Lower middle income	LMC
Low & middle income	LMY
Middle East & North Africa	MEA
Middle income	MIC
Middle East & North Africa (excluding high income)	MNA
North America	NAC
Pacific island small states	PSS
South Asia	SAS
Sub-Saharan Africa (excluding high income)	SSA
Sub-Saharan Africa	SSF
East Asia & Pacific (IDA & IBRD countries)	TEA
Europe & Central Asia (IDA & IBRD countries)	TEC
Latin America & the Caribbean (IDA & IBRD countries)	TLA
Middle East & North Africa (IDA & IBRD countries)	TMN
South Asia (IDA & IBRD)	TSA
Sub-Saharan Africa (IDA & IBRD countries)	TSS
Upper middle income	UMC
World	WLD
South Africa	ZAF

The growth data set contains the same inconsistencies containing information about conglomerates and not just countires as well.

```
growth_raw |>
  filter(
    str_detect(
      `Country Name`,
```

```
      "World|income|IDA|IBRD|Africa|Asia|America|Europe|Caribbean|Pacific|Middle East"
    )
  ) |>
  select(`Country Name`, `Country Code`) |>
  print(n = 40)
```

```
# A tibble: 40 × 2
   `Country Name`                                    `Country Code`
   <chr>                                             <chr>
 1 Africa Eastern and Southern                       AFE
 2 Africa Western and Central                        AFW
 3 Arab World                                        ARB
 4 American Samoa                                    ASM
 5 Central African Republic                          CAF
 6 Central Europe and the Baltics                    CEB
 7 Caribbean small states                            CSS
 8 East Asia & Pacific (excluding high income)       EAP
 9 East Asia & Pacific                               EAS
10 Europe & Central Asia (excluding high income)     ECA
11 Europe & Central Asia                             ECS
12 European Union                                    EUU
13 High income                                       HIC
14 IBRD only                                         IBD
15 IDA & IBRD total                                  IBT
16 IDA total                                         IDA
17 IDA blend                                         IDB
18 IDA only                                          IDX
19 Latin America & Caribbean (excluding high income) LAC
20 Latin America & Caribbean                         LCN
21 Low income                                        LIC
22 Lower middle income                               LMC
23 Low & middle income                               LMY
24 Middle East & North Africa                        MEA
25 Middle income                                     MIC
26 Middle East & North Africa (excluding high income) MNA
27 North America                                     NAC
28 Pacific island small states                       PSS
29 South Asia                                        SAS
30 Sub-Saharan Africa (excluding high income)        SSA
31 Sub-Saharan Africa                                SSF
32 East Asia & Pacific (IDA & IBRD countries)        TEA
33 Europe & Central Asia (IDA & IBRD countries)      TEC
34 Latin America & the Caribbean (IDA & IBRD countries) TLA
35 Middle East & North Africa (IDA & IBRD countries) TMN
36 South Asia (IDA & IBRD)                           TSA
37 Sub-Saharan Africa (IDA & IBRD countries)         TSS
38 Upper middle income                               UMC
39 World                                             WLD
40 South Africa                                      ZAF
```

After listing these entries out we see there are 37 invalid entries.

```
growth_raw |>
  count(`Country Code`) |>
  filter(n > 1)
```

Country Code n

```
#count the number of unique rows we have
growth_raw |>
  summarise(
    rows = n(),
    unique_codes = n_distinct(`Country Code`)
  )
```

rows	unique_codes
266	266

We have identified the main data cleaning problems we will have to complete.

5.2 Assessing Column names

```
names(debt_raw)
```

```
 [1] "DEBT (% of GDP)" "1800"            "1801"            "1802"
 [5] "1803"            "1804"            "1805"            "1806"
 [9] "1807"            "1808"            "1809"            "1810"
[13] "1811"            "1812"            "1813"            "1814"
[17] "1815"            "1816"            "1817"            "1818"
[21] "1819"            "1820"            "1821"            "1822"
[25] "1823"            "1824"            "1825"            "1826"
[29] "1827"            "1828"            "1829"            "1830"
[33] "1831"            "1832"            "1833"            "1834"
[37] "1835"            "1836"            "1837"            "1838"
[41] "1839"            "1840"            "1841"            "1842"
[45] "1843"            "1844"            "1845"            "1846"
[49] "1847"            "1848"            "1849"            "1850"
[53] "1851"            "1852"            "1853"            "1854"
[57] "1855"            "1856"            "1857"            "1858"
[61] "1859"            "1860"            "1861"            "1862"
[65] "1863"            "1864"            "1865"            "1866"
[69] "1867"            "1868"            "1869"            "1870"
[73] "1871"            "1872"            "1873"            "1874"
[77] "1875"            "1876"            "1877"            "1878"
[81] "1879"            "1880"            "1881"            "1882"
[85] "1883"            "1884"            "1885"            "1886"
[89] "1887"            "1888"            "1889"            "1890"
```

```
 [93] "1891"           "1892"           "1893"           "1894"
 [97] "1895"           "1896"           "1897"           "1898"
[101] "1899"           "1900"           "1901"           "1902"
[105] "1903"           "1904"           "1905"           "1906"
[109] "1907"           "1908"           "1909"           "1910"
[113] "1911"           "1912"           "1913"           "1914"
[117] "1915"           "1916"           "1917"           "1918"
[121] "1919"           "1920"           "1921"           "1922"
[125] "1923"           "1924"           "1925"           "1926"
[129] "1927"           "1928"           "1929"           "1930"
[133] "1931"           "1932"           "1933"           "1934"
[137] "1935"           "1936"           "1937"           "1938"
[141] "1939"           "1940"           "1941"           "1942"
[145] "1943"           "1944"           "1945"           "1946"
[149] "1947"           "1948"           "1949"           "1950"
[153] "1951"           "1952"           "1953"           "1954"
[157] "1955"           "1956"           "1957"           "1958"
[161] "1959"           "1960"           "1961"           "1962"
[165] "1963"           "1964"           "1965"           "1966"
[169] "1967"           "1968"           "1969"           "1970"
[173] "1971"           "1972"           "1973"           "1974"
[177] "1975"           "1976"           "1977"           "1978"
[181] "1979"           "1980"           "1981"           "1982"
[185] "1983"           "1984"           "1985"           "1986"
[189] "1987"           "1988"           "1989"           "1990"
[193] "1991"           "1992"           "1993"           "1994"
[197] "1995"           "1996"           "1997"           "1998"
[201] "1999"           "2000"           "2001"           "2002"
[205] "2003"           "2004"           "2005"           "2006"
[209] "2007"           "2008"           "2009"           "2010"
[213] "2011"           "2012"           "2013"           "2014"
[217] "2015"
```

```
names(growth_raw)
```

```
 [1] "Country Name"   "Country Code"   "Indicator Name" "Indicator Code"
 [5] "1960"           "1961"           "1962"           "1963"
 [9] "1964"           "1965"           "1966"           "1967"
[13] "1968"           "1969"           "1970"           "1971"
[17] "1972"           "1973"           "1974"           "1975"
[21] "1976"           "1977"           "1978"           "1979"
[25] "1980"           "1981"           "1982"           "1983"
[29] "1984"           "1985"           "1986"           "1987"
[33] "1988"           "1989"           "1990"           "1991"
[37] "1992"           "1993"           "1994"           "1995"
[41] "1996"           "1997"           "1998"           "1999"
[45] "2000"           "2001"           "2002"           "2003"
[49] "2004"           "2005"           "2006"           "2007"
[53] "2008"           "2009"           "2010"           "2011"
[57] "2012"           "2013"           "2014"           "2015"
```

```
[61] "2016"           "2017"           "2018"           "2019"
[65] "2020"           "2021"           "...67"
```

5.3 Assessing variable type

```
str(debt_raw)
```

```
tibble [197 × 217] (S3: tbl_df/tbl/data.frame)
 $ DEBT (% of GDP): chr [1:197] NA "Afghanistan" "Albania" "Algeria" ...
 $ 1800           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1801           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1802           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1803           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1804           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1805           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1806           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1807           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1808           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1809           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1810           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1811           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1812           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1813           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1814           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1815           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1816           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1817           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1818           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1819           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1820           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1821           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1822           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1823           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1824           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1825           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1826           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1827           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1828           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1829           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1830           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1831           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1832           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1833           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1834           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1835           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1836           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1837           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1838           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1839           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1840           : chr [1:197] NA "no data" "no data" "no data" ...
```

```
$ 1841          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1842          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1843          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1844          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1845          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1846          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1847          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1848          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1849          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1850          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1851          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1852          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1853          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1854          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1855          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1856          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1857          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1858          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1859          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1860          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1861          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1862          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1863          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1864          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1865          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1866          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1867          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1868          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1869          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1870          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1871          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1872          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1873          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1874          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1875          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1876          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1877          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1878          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1879          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1880          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1881          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1882          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1883          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1884          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1885          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1886          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1887          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1888          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1889          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1890          : chr [1:197] NA "no data" "no data" "no data" ...
$ 1891          : chr [1:197] NA "no data" "no data" "no data" ...
```

```
 $ 1892           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1893           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1894           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1895           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1896           : chr [1:197] NA "no data" "no data" "no data" ...
 $ 1897           : chr [1:197] NA "no data" "no data" "no data" ...
  [list output truncated]
```

```
str(growth_raw)
```

```
spc_tbl_ [266 × 67] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Country Name  : chr [1:266] "Aruba" "Africa Eastern and Southern" "Afghanistan" "Africa
Western and Central" ...
 $ Country Code  : chr [1:266] "ABW" "AFE" "AFG" "AFW" ...
 $ Indicator Name: chr [1:266] "GDP growth (annual %)" "GDP growth (annual %)" "GDP growth
(annual %)" "GDP growth (annual %)" ...
 $ Indicator Code: chr [1:266] "NY.GDP.MKTP.KD.ZG" "NY.GDP.MKTP.KD.ZG" "NY.GDP.MKTP.KD.ZG"
"NY.GDP.MKTP.KD.ZG" ...
 $ 1960          : logi [1:266] NA NA NA NA NA NA ...
 $ 1961          : num [1:266] NA 0.24 NA 1.85 NA ...
 $ 1962          : num [1:266] NA 7.98 NA 3.77 NA ...
 $ 1963          : num [1:266] NA 5.16 NA 7.27 NA ...
 $ 1964          : num [1:266] NA 4.58 NA 5.4 NA ...
 $ 1965          : num [1:266] NA 5.33 NA 4.05 NA ...
 $ 1966          : num [1:266] NA 3.91 NA -1.79 NA ...
 $ 1967          : num [1:266] NA 5.26 NA -9.55 NA ...
 $ 1968          : num [1:266] NA 4.02 NA 1.47 NA ...
 $ 1969          : num [1:266] NA 5.28 NA 15.49 NA ...
 $ 1970          : num [1:266] NA 4.7 NA 17.9 NA ...
 $ 1971          : num [1:266] NA 5.37 NA 10.81 NA ...
 $ 1972          : num [1:266] NA 2.15 NA 3.15 NA ...
 $ 1973          : num [1:266] NA 4.44 NA 4.21 NA ...
 $ 1974          : num [1:266] NA 5.89 NA 10.25 NA ...
 $ 1975          : num [1:266] NA 1.73 NA -2.16 NA ...
 $ 1976          : num [1:266] NA 2.85 NA 8.58 NA ...
 $ 1977          : num [1:266] NA 1.23 NA 4.73 NA ...
 $ 1978          : num [1:266] NA 1.03 NA -2.19 NA ...
 $ 1979          : num [1:266] NA 2.8 NA 5.23 NA ...
 $ 1980          : num [1:266] NA 5.42 NA 2 NA ...
 $ 1981          : num [1:266] NA 4.33 NA -6.93 -4.4 ...
 $ 1982          : num [1:266] NA 0.508 NA -3.35 0 ...
 $ 1983          : num [1:266] NA 0.152 NA -6.534 4.2 ...
 $ 1984          : num [1:266] NA 3.006 NA 0.559 6 ...
 $ 1985          : num [1:266] NA -0.448 NA 5.552 3.5 ...
 $ 1986          : num [1:266] NA 2.29 NA 1.41 2.9 ...
 $ 1987          : num [1:266] 16.08 4.23 NA 1.45 4.08 ...
 $ 1988          : num [1:266] 18.65 4 NA 4.84 6.13 ...
 $ 1989          : num [1:266] 12.1298 2.8976 NA 2.2721 0.0416 ...
 $ 1990          : num [1:266] 3.9614 -0.0397 NA 6.5629 -3.4501 ...
 $ 1991          : num [1:266] 7.963 0.105 NA 1.121 0.991 ...
```

```
$ 1992          : num [1:266] 5.88 -1.98 NA 2.69 -5.84 ...
$ 1993          : num [1:266] 7.308 -0.392 NA -1.16 -23.983 ...
$ 1994          : num [1:266] 8.2 2.03 NA -0.3 1.34 ...
$ 1995          : num [1:266] 2.55 4.29 NA 1.93 15 ...
$ 1996          : num [1:266] 1.19 5.44 NA 4.63 13.54 ...
$ 1997          : num [1:266] 7.05 4.42 NA 4.23 7.27 ...
$ 1998          : num [1:266] 1.99 1.85 NA 3.51 4.69 ...
$ 1999          : num [1:266] 1.24 2.64 NA 1.42 2.18 ...
$ 2000          : num [1:266] 7.62 3.35 NA 3.73 3.05 ...
$ 2001          : num [1:266] 4.2 3.66 NA 5.21 4.21 ...
$ 2002          : num [1:266] -0.958 3.893 NA 9.9 13.666 ...
$ 2003          : num [1:266] 1.12 3.08 8.83 5.52 2.99 ...
$ 2004          : num [1:266] 7.28 5.51 1.41 8.01 10.95 ...
$ 2005          : num [1:266] -0.385 6.117 11.23 5.848 15.03 ...
$ 2006          : num [1:266] 1.14 6.55 5.36 5.37 11.55 ...
$ 2007          : num [1:266] 3.1 6.6 13.83 5.53 14.01 ...
$ 2008          : num [1:266] 1.83 4.34 3.92 6.28 11.17 ...
$ 2009          : num [1:266] -11.684 0.761 21.391 6.274 0.86 ...
$ 2010          : num [1:266] -2.73 5.15 14.36 6.96 4.86 ...
$ 2011          : num [1:266] 3.367 3.681 0.426 4.848 3.47 ...
$ 2012          : num [1:266] -1.035 0.923 12.752 5.143 8.54 ...
$ 2013          : num [1:266] 6.43 4.2 5.6 6.1 4.95 ...
$ 2014          : num [1:266] -0.0205 3.9844 2.7245 5.9274 4.82 ...
$ 2015          : num [1:266] 3.54 2.95 1.45 2.75 0.94 ...
$ 2016          : num [1:266] 2.117 2.217 2.26 0.128 -2.58 ...
$ 2017          : num [1:266] 5.48 2.56 2.65 2.32 -0.15 ...
$ 2018          : num [1:266] 1.27 2.49 1.19 2.95 -1.32 ...
$ 2019          : num [1:266] -2.09 2.03 3.91 3.2 -0.7 ...
$ 2020          : num [1:266] -22.319 -2.887 -2.351 -0.897 -5.5 ...
$ 2021          : num [1:266] NA 4.3 NA 3.91 0.7 ...
$ ...67         : logi [1:266] NA NA NA NA NA NA ...
- attr(*, "spec")=
 .. cols(
 ..   `Country Name` = col_character(),
 ..   `Country Code` = col_character(),
 ..   `Indicator Name` = col_character(),
 ..   `Indicator Code` = col_character(),
 ..   `1960` = col_logical(),
 ..   `1961` = col_double(),
 ..   `1962` = col_double(),
 ..   `1963` = col_double(),
 ..   `1964` = col_double(),
 ..   `1965` = col_double(),
 ..   `1966` = col_double(),
 ..   `1967` = col_double(),
 ..   `1968` = col_double(),
 ..   `1969` = col_double(),
 ..   `1970` = col_double(),
 ..   `1971` = col_double(),
 ..   `1972` = col_double(),
 ..   `1973` = col_double(),
```

```
..   `1974` = col_double(),
..   `1975` = col_double(),
..   `1976` = col_double(),
..   `1977` = col_double(),
..   `1978` = col_double(),
..   `1979` = col_double(),
..   `1980` = col_double(),
..   `1981` = col_double(),
..   `1982` = col_double(),
..   `1983` = col_double(),
..   `1984` = col_double(),
..   `1985` = col_double(),
..   `1986` = col_double(),
..   `1987` = col_double(),
..   `1988` = col_double(),
..   `1989` = col_double(),
..   `1990` = col_double(),
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..   `1994` = col_double(),
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..   `1997` = col_double(),
..   `1998` = col_double(),
..   `1999` = col_double(),
..   `2000` = col_double(),
..   `2001` = col_double(),
..   `2002` = col_double(),
..   `2003` = col_double(),
..   `2004` = col_double(),
..   `2005` = col_double(),
..   `2006` = col_double(),
..   `2007` = col_double(),
..   `2008` = col_double(),
..   `2009` = col_double(),
..   `2010` = col_double(),
..   `2011` = col_double(),
..   `2012` = col_double(),
..   `2013` = col_double(),
..   `2014` = col_double(),
..   `2015` = col_double(),
..   `2016` = col_double(),
..   `2017` = col_double(),
..   `2018` = col_double(),
..   `2019` = col_double(),
..   `2020` = col_double(),
..   `2021` = col_double(),
..   ...67 = col_logical()
```

```
  .. )
- attr(*, "problems")=<externalptr>
```

5.4 Assessing data completeness

```
debt_raw |>
  summarise(
    n_rows = n(),
    n_countries = n_distinct(`DEBT (% of GDP)`)
  )
```

n_rows	n_countries
197	196

The number of rows in not the same value as the number of countries. double checking yields

```
debt_raw |>
  slice(2:190) |>
  summarise(
    n_rows = n(),
    n_countries = n_distinct(`DEBT (% of GDP)`)
  )
```

n_rows	n_countries
189	189

Doing the same for the growth data

```
growth_raw |>
  summarise(
    n_rows = n(),
    n_entities = n_distinct(`Country Code`)
  )
```

n_rows	n_entities
266	266

completeness check

```
growth_raw |>
  filter(
    !`Country Code` %in% c(
      "AFE", "AFW", "ARB", "CEB", "CSS",
      "EAP", "EAS", "ECA", "ECS", "EUU",
      "HIC", "IBD", "IBT", "IDA", "IDB", "IDX",
      "LAC", "LCN", "LIC", "LMC", "LMY",
```

```
      "MEA", "MIC", "MNA", "NAC", "PSS",
      "SAS", "SSA", "SSF", "TEA", "TEC",
      "TLA", "TMN", "TSA", "TSS", "UMC", "WLD"
    )
  ) |>
  summarise(
    n_rows = n(),
    n_countries = n_distinct(`Country Code`)
  )
```

n_rows	n_countries
229	229

After we exclude the conglomerates and isolate individual countries we have 229 countries.

6 Step 4: Clean and pre-process the data

As seen in our exploration steps we must remove the non countries form the data sets and isolate the individual countries with no duplicates.

```
#create a clean_debt function to manually exclude row 1
#NOT overriding the original file
clean_debt <- function(debt_raw) {

  debt_clean <- debt_raw |>
    slice(2:190) |>
    select(`DEBT (% of GDP)`, `1960`:`2015`)

  debt_clean
}
```

```
debt_clean <- clean_debt(debt_raw)
```

```
#sanity check
dim(debt_clean)
```

```
[1] 189  57
```

Now converting "no data" -> NA

```
clean_debt_missing <- function(debt_clean) {
  debt_clean <- debt_clean |>
    mutate(
      across(
        `1960`:`2015`,
        ~ na_if(.x, "no data")
```

```
      )
    )

  debt_clean
}
```

```
#sanity check
debt_clean2 <- clean_debt_missing(debt_clean)
sum(is.na(debt_clean2))
```

```
[1] 3354
```

Then we make all values numeric

```
clean_debt_numeric <- function(debt_clean) {

  debt_clean <- debt_clean |>
    mutate(
      across(
        `1960`:`2015`,
        as.numeric
      )
    )

  debt_clean
}

debt_clean3 <- clean_debt_numeric(debt_clean2)
```

```
#sanity check
str(debt_clean3)
```

```
tibble [189 × 57] (S3: tbl_df/tbl/data.frame)
 $ DEBT (% of GDP): chr [1:189] "Afghanistan" "Albania" "Algeria" "Angola" ...
 $ 1960           : num [1:189] NA NA NA NA NA ...
 $ 1961           : num [1:189] NA NA NA NA NA ...
 $ 1962           : num [1:189] NA NA NA NA NA ...
 $ 1963           : num [1:189] NA NA NA NA NA ...
 $ 1964           : num [1:189] NA NA NA NA NA ...
 $ 1965           : num [1:189] NA NA NA NA NA NA NA NA NA NA ...
 $ 1966           : num [1:189] NA NA NA NA NA ...
 $ 1967           : num [1:189] NA NA NA NA NA ...
 $ 1968           : num [1:189] NA NA NA NA NA ...
 $ 1969           : num [1:189] NA NA NA NA NA ...
 $ 1970           : num [1:189] NA NA 27 NA NA ...
 $ 1971           : num [1:189] NA NA 31.8 NA NA ...
 $ 1972           : num [1:189] NA NA 27.5 NA NA ...
 $ 1973           : num [1:189] NA NA 33.4 NA NA ...
 $ 1974           : num [1:189] NA NA 32.5 NA NA ...
```

```
$ 1975           : num [1:189] NA NA 36.6 NA NA ...
$ 1976           : num [1:189] NA NA 43.3 NA NA ...
$ 1977           : num [1:189] NA NA 54.5 NA NA ...
$ 1978           : num [1:189] NA NA 70.3 NA NA ...
$ 1979           : num [1:189] NA NA 65.2 NA NA ...
$ 1980           : num [1:189] NA NA 55.6 NA NA ...
$ 1981           : num [1:189] NA NA 44.7 NA NA ...
$ 1982           : num [1:189] NA NA 44.5 NA NA ...
$ 1983           : num [1:189] NA NA 46.8 NA NA ...
$ 1984           : num [1:189] NA NA 47.3 NA NA ...
$ 1985           : num [1:189] NA NA 47 NA NA ...
$ 1986           : num [1:189] NA NA 60.5 NA NA ...
$ 1987           : num [1:189] NA NA 69.3 NA NA ...
$ 1988           : num [1:189] NA NA 83.4 NA NA ...
$ 1989           : num [1:189] NA NA 78.4 NA NA ...
$ 1990           : num [1:189] NA NA 69 NA NA ...
$ 1991           : num [1:189] NA NA 71.7 NA NA ...
$ 1992           : num [1:189] NA NA 70.1 NA NA ...
$ 1993           : num [1:189] NA NA 97.5 NA NA ...
$ 1994           : num [1:189] NA 85.2 98 NA NA ...
$ 1995           : num [1:189] NA 75 94.3 222.1 NA ...
$ 1996           : num [1:189] NA 76.7 79.2 147.9 NA ...
$ 1997           : num [1:189] NA 81.9 75.8 118.1 NA ...
$ 1998           : num [1:189] NA 71.9 79.8 154.9 NA ...
$ 1999           : num [1:189] NA 63.9 72.2 165.2 NA ...
$ 2000           : num [1:189] NA 64.6 61.25 89.9 9.13 ...
$ 2001           : num [1:189] NA 60.6 59.8 80.2 10.6 ...
$ 2002           : num [1:189] 281 64.1 59.4 73 12.2 ...
$ 2003           : num [1:189] 273.5 60.2 50.6 59.8 11.7 ...
$ 2004           : num [1:189] 243.4 57.5 39.3 52 16 ...
$ 2005           : num [1:189] 204 58.2 28.6 35.7 17.4 ...
$ 2006           : num [1:189] 23 56.7 24.3 18.7 14 ...
$ 2007           : num [1:189] 20.2 53.5 13.6 16.1 14 ...
$ 2008           : num [1:189] 18.37 55.14 8.39 16.65 15.4 ...
$ 2009           : num [1:189] 16.8 59.7 10.2 22.7 25.1 ...
$ 2010           : num [1:189] 7.9 57.7 10.9 44.3 32.5 ...
$ 2011           : num [1:189] 7.14 59.41 9.49 33.8 29.21 ...
$ 2012           : num [1:189] 6.63 62.14 9.49 29.49 30.94 ...
$ 2013           : num [1:189] 6.75 70.39 7.74 32.87 30.42 ...
$ 2014           : num [1:189] 6.38 72.04 7.96 40.66 24.11 ...
$ 2015           : num [1:189] 6.25 73.32 9.06 64.24 22.15 ...
```

Next to match the given format we must change the format of the dataframe into a long and skinny matrix rather than a short and fat matrix.

```
clean_debt_long <- function(debt_clean) {

  debt_clean <- debt_clean |>
    pivot_longer(
      cols = `1960`:`2015`,
```

```
      names_to = "year",
      values_to = "debt_pct_gdp"
    ) |>
    rename(
      country = `DEBT (% of GDP)`
    )

  debt_clean
}

debt_clean4 <- clean_debt_long(debt_clean3)
```

```
#check
dim(debt_clean4)
```

```
[1] 10584     3
```

```
head(debt_clean4)
```

country	year	debt_pct_gdp
Afghanistan	1960	NA
Afghanistan	1961	NA
Afghanistan	1962	NA
Afghanistan	1963	NA
Afghanistan	1964	NA
Afghanistan	1965	NA

```
str(debt_clean4)
```

```
tibble [10,584 × 3] (S3: tbl_df/tbl/data.frame)
 $ country     : chr [1:10584] "Afghanistan" "Afghanistan" "Afghanistan" "Afghanistan" ...
 $ year        : chr [1:10584] "1960" "1961" "1962" "1963" ...
 $ debt_pct_gdp: num [1:10584] NA NA NA NA NA NA NA NA NA NA ...
```

All functions worked properly but year is still not numeric.

```
#make year numeric
clean_debt_year <- function(debt_clean) {

  debt_clean <- debt_clean |>
    mutate(
      year = as.integer(year)
    )

  debt_clean
```

```
}

debt_clean5 <- clean_debt_year(debt_clean4)
```

```
str(debt_clean5)
```

```
tibble [10,584 × 3] (S3: tbl_df/tbl/data.frame)
 $ country     : chr [1:10584] "Afghanistan" "Afghanistan" "Afghanistan" "Afghanistan" ...
 $ year        : int [1:10584] 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 ...
 $ debt_pct_gdp: num [1:10584] NA NA NA NA NA NA NA NA NA NA ...
```

The debt data frame is prepared for shape manipulation, shifting attention to growth.csv

We know we must remove the conglomerates

```
clean_growth <- function(growth_raw) {

  growth_clean <- growth_raw |>
    select(-`...67`) |>
    filter(
      !`Country Code` %in% c(
        "AFE", "AFW", "ARB", "CEB", "CSS",
        "EAP", "EAS", "ECA", "ECS", "EUU",
        "HIC", "IBD", "IBT", "IDA", "IDB", "IDX",
        "LAC", "LCN", "LIC", "LMC", "LMY",
        "MEA", "MIC", "MNA", "NAC", "PSS",
        "SAS", "SSA", "SSF", "TEA", "TEC",
        "TLA", "TMN", "TSA", "TSS", "UMC", "WLD"
      )
    )

  growth_clean
}
growth_clean <- clean_growth(growth_raw)
```

```
#check
dim(growth_clean)
```

```
[1] 229  66
```

```
#truncate data to 1960:2021
clean_growth_columns <- function(growth_clean) {

  growth_clean <- growth_clean |>
    select(
      `Country Name`,
      `1960`:`2021`
    )
```

```
  growth_clean
}
growth_clean2 <- clean_growth_columns(growth_clean)
```

```
#check
dim(growth_clean2)
```

```
[1] 229  63
```

We are now ready to mutate the matrices

```
clean_growth_long <- function(growth_clean) {

  growth_clean <- growth_clean |>
    pivot_longer(
      cols = `1960`:`2021`,
      names_to = "year",
      values_to = "growth_pct_gdp"
    ) |>
    rename(
      country = `Country Name`
    )

  growth_clean
}
growth_clean3 <- clean_growth_long(growth_clean2)
```

```
#check
dim(growth_clean3)
```

```
[1] 14198     3
```

```
head(growth_clean3)
```

country	year	growth_pct_gdp
Aruba	1960	NA
Aruba	1961	NA
Aruba	1962	NA
Aruba	1963	NA
Aruba	1964	NA
Aruba	1965	NA

```
str(growth_clean3)
```

```
tibble [14,198 × 3] (S3: tbl_df/tbl/data.frame)
 $ country       : chr [1:14198] "Aruba" "Aruba" "Aruba" "Aruba" ...
 $ year          : chr [1:14198] "1960" "1961" "1962" "1963" ...
 $ growth_pct_gdp: num [1:14198] NA NA NA NA NA NA NA NA NA NA ...
```

```r
#change year to numeric
clean_growth_year <- function(growth_clean) {

  growth_clean <- growth_clean |>
    mutate(
      year = as.integer(year)
    )

  growth_clean
}

growth_clean4 <- clean_growth_year(growth_clean3)
```

```r
str(growth_clean4)
```

```
tibble [14,198 × 3] (S3: tbl_df/tbl/data.frame)
 $ country       : chr [1:14198] "Aruba" "Aruba" "Aruba" "Aruba" ...
 $ year          : int [1:14198] 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 ...
 $ growth_pct_gdp: num [1:14198] NA NA NA NA NA NA NA NA NA NA ...
```

And a sanity check

```r
#sanity check
dim(growth_clean4)
```

```
[1] 14198     3
```

```r
summary(growth_clean4)
```

```
   country               year      growth_pct_gdp
 Length:14198       Min.   :1960   Min.   :-64.047
 Class :character   1st Qu.:1975   1st Qu.:  1.300
 Mode  :character   Median :1990   Median :  3.786
                    Mean   :1990   Mean   :  3.647
                    3rd Qu.:2006   3rd Qu.:  6.200
                    Max.   :2021   Max.   :149.973
                                   NA's   :3836
```

```r
head(growth_clean4)
```

country	year	growth_pct_gdp
Aruba	1960	NA

country	year	growth_pct_gdp
Aruba	1961	NA
Aruba	1962	NA
Aruba	1963	NA
Aruba	1964	NA
Aruba	1965	NA

Now we are all set to inner joing the two tables

```
preprocess_data <- function(debt_clean, growth_clean) {

  combined <- debt_clean |>
    inner_join(
      growth_clean,
      by = c("country", "year")
    )

  combined
}
combined_data <- preprocess_data(
  debt_clean5,
  growth_clean4
)
```

```
#sanity check
dim(combined_data)
```

```
[1] 9352    4
```

```
#preview combined table
head(combined_data)
```

country	year	debt_pct_gdp	growth_pct_gdp
Afghanistan	1960	NA	NA
Afghanistan	1961	NA	NA
Afghanistan	1962	NA	NA
Afghanistan	1963	NA	NA
Afghanistan	1964	NA	NA
Afghanistan	1965	NA	NA

```
summary(combined_data)
```

```
  country               year       debt_pct_gdp     growth_pct_gdp
Length:9352        Min.   :1960   Min.   :   0.00   Min.   :-64.047
Class :character   1st Qu.:1974   1st Qu.:  24.67   1st Qu.:  1.450
Mode  :character   Median :1988   Median :  43.33   Median :  3.927
                   Mean   :1988   Mean   :  55.74   Mean   :  3.882
                   3rd Qu.:2001   3rd Qu.:  68.94   3rd Qu.:  6.328
                   Max.   :2015   Max.   :2092.92   Max.   :149.973
                                  NA's   :2851      NA's   :2237
```